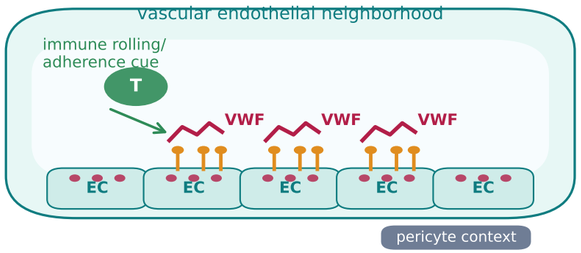
pyXenium identifies a topology-supported endothelial VWF-SELP vascular activation niche

Whole-dataset Xenium WTA breast analysis: rank 1 LR axis, endothelial-specific expression, spatial hotspots, S3 vascular contour enrichment, and cross-method support.

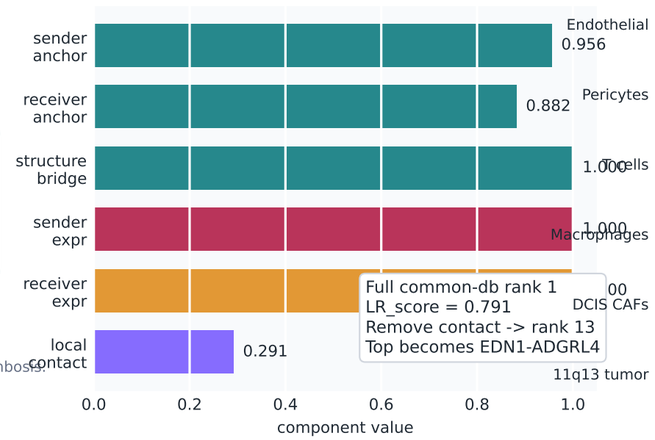
A Endothelial activation model

VWF + SELP/P-selectin = WPB / adhesion state

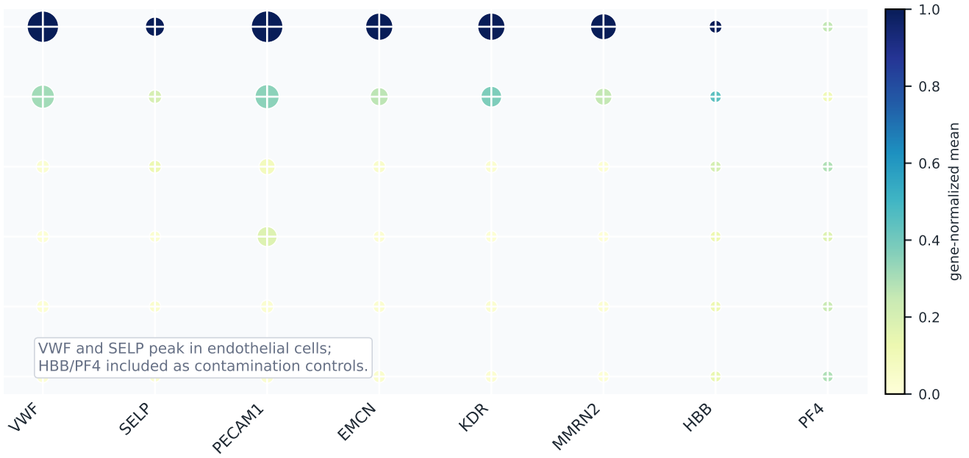


Caveat: RNA + spatial evidence; not direct proof of protein release or thrombosis.

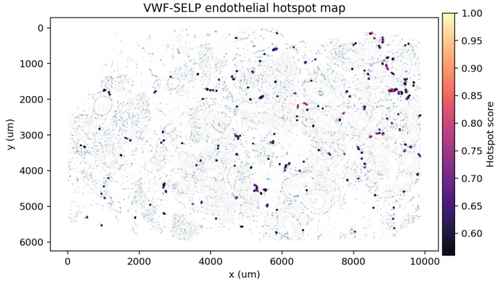
B pyXenium score stack



C WTA endothelial specificity

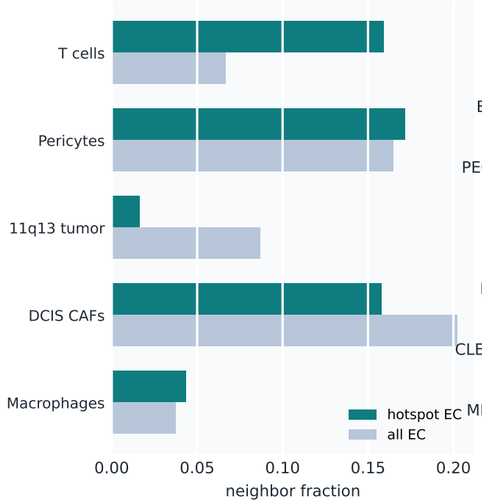


D Spatial hotspots and neighborhood context

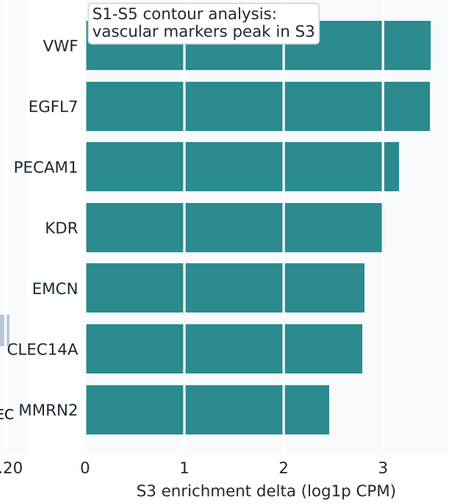


433 endothelial hotspot cells
95th percentile among endothelial cells

hotspots are immune/pericyte-associated, not simply tumor-front enriched



E Tissue ecology and method triangulation



Exact VWF-SELP recovered by other methods

method	rank	score_std
pyxenium	1	1.000
cellphonedb	692	0.999
laris	2,845	0.998
liana	8,450	0.989

Interpretation: consensus supports a vascular axis; pyXenium adds topology/contact specificity.

Main interpretation: the top pyXenium hit is best read as an endothelial activation / Weibel-Palade body-associated adhesion state. The data support RNA-level endothelial specificity and spatial topology; protein release, platelet adhesion, and thrombosis remain biological hypotheses requiring orthogonal validation.